

Team Worksheet — Lab 26

FMEA, Pilot Design and Implementation Planning — In Depth

Team members: _____

Date: _____

1. Our understanding of the problem

2. Our working

Use the data in lab-26-fmea-worksheet.xlsx.

3. What we produced

Deliverable: A completed FMEA with re-scored RPNs and a full pilot plan with success criteria.

4. What the evidence does NOT tell us

5. Our conclusion

6. Step notes

Step 1. Recap FMEA in full: failure mode, effect, cause, current controls, and the three ratings that combine into the risk priority number.

Step 2. Recap the scoring scales precisely: Severity (impact if it occurs), Occurrence (likelihood) and Detection (probability of catching it BEFORE the customer does — note the inverted scale).

Step 3. Build the FMEA for your selected improvement. For each process step, ask what could fail, what happens if it does, and what would cause it.

Step 4. Score S, O and D on 1-10 scales and calculate $RPN = S \times O \times D$.

Step 5. Apply the severity override rule: any failure mode with Severity 9-10 requires action regardless of how low its RPN is.

Step 6. Rank by RPN and define mitigation for every high-RPN mode, targeting the component you can most cheaply move — usually Detection, sometimes Occurrence.

Step 7. RE-SCORE after mitigation and record the residual RPN. An FMEA without re-scoring has not demonstrated that risk was actually reduced.

Step 8. Design the pilot: scope (which line, site or segment), duration, sample size and the exact data to be collected.

Step 9. Define quantitative success criteria BEFORE the pilot starts, expressed against your Lab 3 baseline. Criteria defined afterwards get rationalised.

Step 10. Plan the comparison: before/after, or better, a control group running the unchanged process concurrently to rule out external effects.

Step 11. Define the stop conditions — what result would cause you to halt the pilot rather than push through.

Step 12. Build the implementation plan: tasks, owners, dates, training, communication, systems changes and documentation updates.

Step 13. Complete the cost-benefit analysis: implementation cost, annualised benefit, payback period and NPV where the investment is material.

Step 14. Run the pilot (or design it fully for post-course execution) and test the results statistically against the baseline — a visual improvement is not proof.

7. Self-check

Your FMEA is re-scored post-mitigation with the severity override applied, pilot success criteria were defined before starting, and the improvement is tested statistically against the documented baseline.

Complete? ☐ Yes ☐ Not yet

Certified Lean Six Sigma Black Belt (CLSSBB) Training · TGS-2024051900 · Version v4 · © 2026 Tertiary Infotech Academy Pte Ltd